

Yousif Abdoun

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Objective

Electrical Engineering student specializing in low-latency systems, computer architecture, and embedded platforms, with hands-on experience in digital design, performance-critical systems, and real-time hardware–software co-design.

Education

Georgia Institute of Technology | Atlanta, GA

Fall 2028

Bachelor of Science in Electrical Engineering and Mathematics (Double Major), GPA: 3.85/4.00

Skills

Programming: C/ C++, SystemsVerilog, MATLAB, Python, Java,

Platforms / OS: Linux (Ubuntu), ROS

Professional Organizations: Silicon Jackets (Robotics club), Quantitative Finance Club, National Society for Black Engineers

Languages: English (Native), Arabic (Native)

Experience

Intelligent Vision and Automations Laboratory (Vertically Integrated Projects)

Atlanta, Georgia

Undergraduate Student Researcher – Embedded Systems subteam

Jan 2026 – April 2026

Autobots VIP is a Georgia Tech research program focused on designing and deploying autonomous mobile robots using robotics, computer vision, control, and AI to solve real-world, task-driven problems.

- Designed and optimized low-latency, distributed message-passing pipelines in ROS (C++/Python) on Linux, characterizing inter-node communication via latency, throughput, and drop-rate metrics under high-frequency workloads
- Profiled and debugged concurrent multi-process systems using Linux tooling (perf, gdb), identifying synchronization faults and race conditions to improve system stability and reduce failure rates by >40%
- Analyzed compute–communication bottlenecks in real-time perception pipelines, evaluating CPU–GPU data transfer overhead and applying batching/asynchronous execution to improve throughput in heterogeneous systems

Projects

Real-Time OS Scheduler Simulator — C++ (system), Python (Pandas, Matplotlib), EDF/RM Scheduling Algorithms

- Built a preemptive real-time scheduling engine implementing EDF, Rate-Monotonic, and FCFS policies with deadline tracking, preemption counting, and jitter analysis on a simulated single-core CPU.
- Designed a discrete-time kernel simulator that generates periodic task workloads, executes job-level scheduling decisions, and records full CPU execution traces (Gantt timelines).
- Developed a Python analytics pipeline using Pandas and Matplotlib to visualize response times, deadline slack, jitter, and scheduling fairness across competing policies.

TOAD (Terraforming Oxygenation and Agricultural Droid) — C++, Fushion 360, Robotics, Low-Latency Wireless Control

- Designed and built a distributed robotic rover using dual ESP32 microcontrollers communicating via ESP-NOW, achieving 10–20 ms end-to-end control latency for real-time remote operation.
- Implemented coordinated actuator control for a multi-stage planting pipeline (servo-driven soil penetration, chassis alignment, and seed dispensing), integrating 4 DC motors and 2 servos with strict motion constraints to prevent mechanical misalignment.
- Developed embedded C++ firmware using Arduino and VS Code with GitHub version control, while custom-fabricating mechanical components (3D-printed hopper, gear system, mounts, chassis, wheels) under power and resource constraints.

CLIPT — Python, Java

- Built a FastAPI-based Python web service that ingests YouTube/Twitch VODs, orchestrates asynchronous media-processing jobs, and generates fixed-length (30s) highlight clips via a custom hype-scoring pipeline.
- Implemented audio signal–level feature extraction (amplitude envelopes, temporal aggregation, peak detection, rolling statistics) and generated time-aligned heatmaps to localize high-intensity segments for downstream scoring.
- Designed a modular backend architecture spanning ingestion, scoring, caption generation, and video rendering with explicit state/status management and MLflow-backed experiment logging for reproducibility.